

JULIAN DÖRNER, MASTER OF SCIENCE

CONTACT

University:
Englerstr. 2
76131 Karlsruhe
Germany
mail [at] juliandoerner.de

POSITIONS

PhD student in numerical analysis

Since August 2023

Karlsruhe Institute of Technology

Karlsruhe, Germany

- Junior research group: Numerical methods for nonlinear optics
- Supervision by Dr. Benjamin Dörich and Prof. Dr. Roland Schnaubelt
- Numerical analysis for nano- and quantum optics, within the Collaborative Research Centre 1173 – Wave Phenomena
- **Expected completion:** December 2026

Visiting PhD scholar

September 2025 –

French Institute for Research in Computer Science and Automation (Inria)

February 2026

Lille, France

- RAPSODI team, supervision by Dr. Théophile Chaumont-Frelet
- Wave scattering with non-local transmission conditions
- Funded by a Research Travel Grant of the Karlsruhe House of Young Scientists

Student research assistant

October 2020 – July 2023

Karlsruhe Institute of Technology

Karlsruhe, Germany

- Numerical Analysis Group

Student research assistant

March 2020 – September 2020

Karlsruhe Institute of Technology

Karlsruhe, Germany

- Lattice Boltzmann Research Group

EDUCATION

Master of Science (with distinction), Mathematics

October 2021 – June 2023

Karlsruhe Institute of Technology

Karlsruhe, Germany

- Focus on numerics and analysis, minor in computer science
- Thesis: ‘Discontinuous Galerkin Discretization for Maxwell Equations with Inhomogeneous Interface Conditions’

Bachelor of Science, Mathematics

October 2017 – July 2021

Karlsruhe Institute of Technology

Karlsruhe, Germany

- Focus on numerics and analysis, minor in business administration
- Thesis: ‘Stability of Runge–Kutta methods for delay differential equations’

PRIZES AND SCHOLARSHIPS

Congress travel grant

German Academic Exchange Service (DAAD)

March 2026

Germany

Research Travel Grant

Karlsruhe House of Young Scientists

February 2025

Karlsruhe, Germany

- Funding of a 6-month research stay in Lille, France

Prize for an outstanding Master's thesis

Collaborative Research Centre 1173 – Wave Phenomena

December 2023

Karlsruhe, Germany

SKILLS

Programming—C/C++ and Python

FEM Libraries—`deal.II` & `FEniCSx`

HPC Techniques—Basic familiarity with MPI and CUDA

Languages—German (native), English (proficient), French (elementary), Spanish (elementary)